



**CHENNAI
INSTITUTE OF TECHNOLOGY**
(Autonomous)



RTOS Based Water Quality Monitoring System

USER & HANDOVER MANUAL

TRON Programming Contest 2026

QuadCore • EE VLSI

Prepared for hardware handover, reassembly, operation and technical reference

1. Document Control & Team

Item	Details
Project title	RTOS Based Water Quality Monitoring System
Team	QuadCore
Competition	TRON Programming Contest 2026
Institution	Chennai Institute of Technology
Department	EE VLSI
Document purpose	User operation, wiring reference, reassembly, troubleshooting and technical handover
Revision	Rev 1.0
Prepared for	TRON Programming Contest 2026 submission / hardware handover

Team Members

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Document Use

This manual is intended to remain with the hardware after the prototype is dismantled for transport. It therefore records the electrical connections, sensor interfaces, ADC protection network, software/data pipeline, operating sequence, calibration checks, troubleshooting guidance and reassembly notes.

2. Project Overview

The RTOS Based Water Quality Monitoring System measures three water-quality parameters—pH, total dissolved solids (TDS), and turbidity—using analog sensor modules connected to a Nucleo-based controller. The measured values are processed by embedded firmware and used with a TensorFlow-trained autoencoder model for anomaly detection.

Primary Measurements

Parameter	Sensor signal	Controller input	Purpose
pH	Analog voltage	A0 via 10 kΩ / 10 kΩ divider	Acidity/alkalinity measurement
TDS	Analog output	A1	Dissolved-solids estimation
Turbidity	Analog output	A5	Water-cloudiness / suspended-particle indication

System-Level Flow

Water sample → Sensors → Analog conditioning → Nucleo ADC → RTOS firmware → normalization → autoencoder → anomaly decision → serial/console output

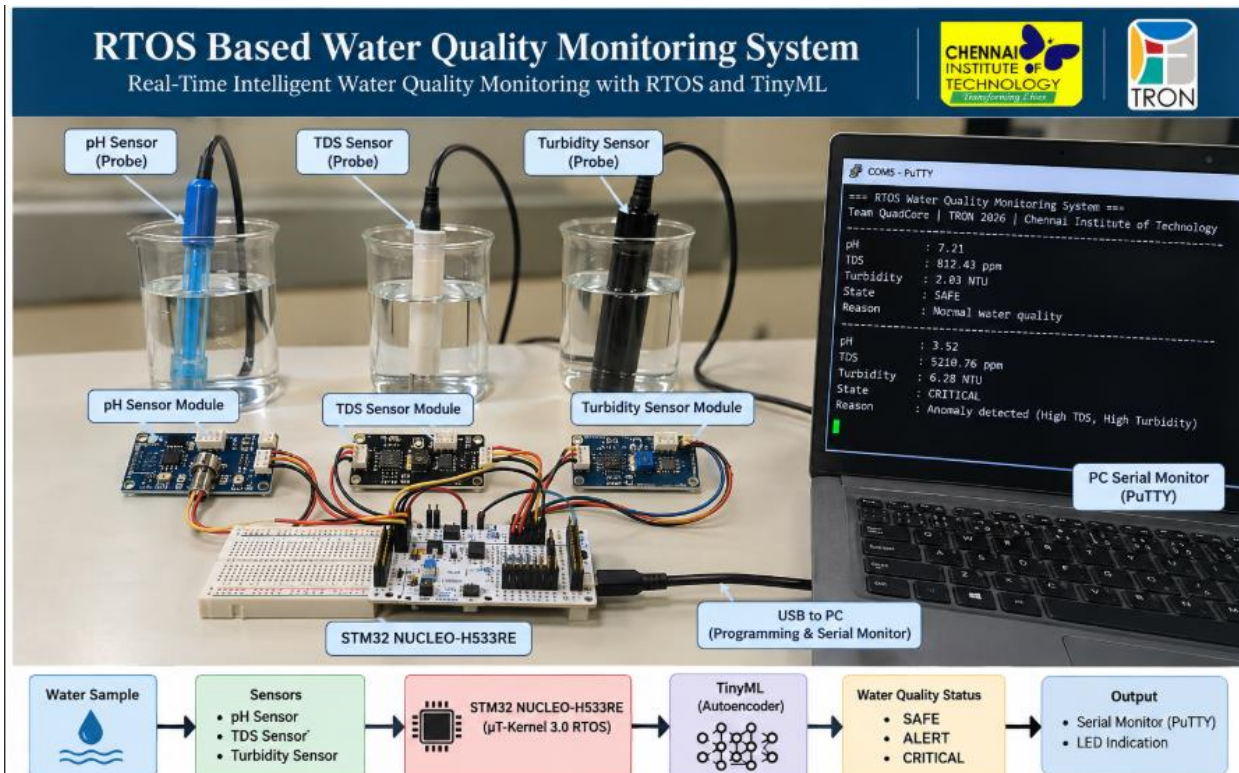


Figure 1: Complete Prototype Setup of the RTOS Based Water Quality Monitoring System

3. System Architecture

Functional Architecture

Layer	Function
Sensing layer	pH, TDS and turbidity modules acquire physical water-quality information.
Signal-conditioning layer	pH output is attenuated using R1/R2 before entering the 3.3 V-class ADC input.
Acquisition layer	Nucleo ADC channels sample A0, A1 and A5.
RTOS / firmware layer	Coordinates periodic acquisition and processing of sensor data.
Pre-processing layer	Applies the documented min-max normalization parameters.
TinyML layer	Runs the embedded autoencoder inference.
Decision layer	Compares reconstruction error against the anomaly threshold.
Interface layer	Reports measurements and model results through the configured serial/debug interface.

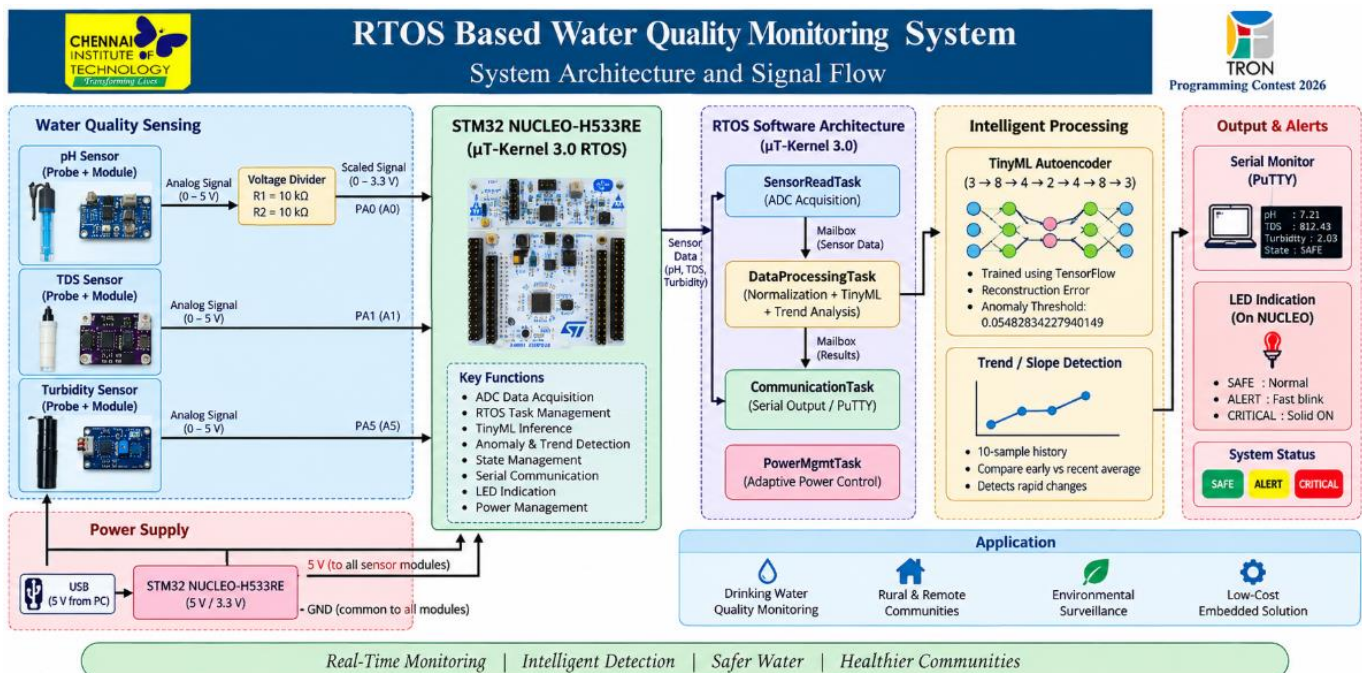
Logical Signal Path

pH → R1 (10 kΩ) → ADC node A0 → R2 (10 kΩ) → GND

TDS → A1

Turbidity → A5

All module grounds → common Nucleo GND



4. Hardware Connections & Pinout

The following wiring reflects the connection information supplied for the final prototype.

Module / Signal	Connection	Notes
pH VCC	Nucleo 5V	Module supply
pH GND	Nucleo common GND	Shared ground
pH OUT	R1 = 10 k Ω $\rightarrow$ ADC node	Do not bypass the divider
ADC node	Nucleo A0	Node between R1 and R2
R2 lower end	Nucleo GND	R2 = 10 k Ω
TDS VCC	Nucleo 5V	Module supply
TDS GND	Nucleo common GND	Shared ground
TDS analog OUT	Nucleo A1	Analog measurement input
Turbidity VCC	Nucleo 5V	Module supply
Turbidity GND	Nucleo common GND	Shared ground
Turbidity analog OUT	Nucleo A5	Analog measurement input

Common Power Reference

- Nucleo 5V rail supplies all three sensor modules.
- All three module grounds are tied to the Nucleo common ground.
- The pH signal uses a dedicated two-resistor divider before A0.
- Keep the analog signal wires physically separated from noisy power or switching wires where practical.

5. pH ADC Voltage-Divider Connection

The pH module can present a voltage above the preferred ADC input range. The prototype therefore uses a 1:1 resistive divider made from two 10 kΩ resistors.

Exact Connection

Component	From	To	Value
R1	pH module OUT	ADC node / A0	10 kΩ
R2	ADC node / A0	GND	10 kΩ

Electrical Relationship

$$V_{\text{ADC}} = V_{\text{pH}} \times R2 / (R1 + R2)$$

$$\text{With } R1 = R2 = 10 \text{ k}\Omega: V_{\text{ADC}} \approx V_{\text{pH}} / 2$$

Example: if the pH module output is approximately 3.96 V, the ideal divider node is approximately 1.98 V. The exact measured value can differ slightly due to resistor tolerance, wiring, sensor output variation and meter/ADC loading.

Critical Reassembly Check

1. Locate the pH module OUT wire.
2. Connect OUT to the upper end of R1 (10 kΩ).
3. Connect the junction of R1 and R2 to Nucleo A0.
4. Connect the lower end of R2 (10 kΩ) to the common GND.
5. Confirm that A0 is connected to the resistor junction—not directly to the raw pH output.
6. Before powering the Nucleo, verify the two resistors are actually connected as a divider.

6. Sensor Placement & Measurement Procedure

Before Measurement

7. Place the system on a dry, stable surface.
8. Verify the common ground connection and the three sensor power connections.
9. Verify the pH divider is present and correctly wired.
10. Connect the required Nucleo programming/debug interface.
11. Power the system and allow the sensor electronics to stabilize according to the sensor-module operating procedure.

During Measurement

12. Prepare the water sample in a suitable container.
13. Position the sensing elements according to their individual module/probe requirements.
14. Avoid allowing exposed electronics or the breadboard to contact the water.
15. Wait for the readings to stabilize before recording or comparing values.
16. Observe the serial/debug output for the three sensor values and the anomaly result.

After Measurement

- Remove probes from the sample and follow the sensor manufacturer's cleaning/storage guidance.
- Do not transport the electronics with wet probes attached.
- For shipment, separate the probes and fragile sensor parts and protect them against mechanical shock.

7. RTOS & Firmware Operation

The firmware is structured around an RTOS-based acquisition and processing workflow. The manual records the functional pipeline rather than assuming names for individual RTOS tasks, since task names and scheduling details should match the final source code delivered with the hardware.

Stage	Function	Output
1. Acquisition	Read analog sensor channels A0, A1 and A5.	Raw ADC values / sensor-domain values
2. Conversion	Convert raw readings into the configured pH, TDS and turbidity representations.	Three sensor features
3. Normalization	Scale each feature using the stored minimum and maximum values.	Normalized 3-element vector
4. Inference	Run the embedded autoencoder.	Reconstructed 3-element vector
5. Error calculation	Calculate reconstruction error.	Anomaly score
6. Decision	Compare anomaly score with 0.05482834227940149.	Normal / anomaly indication
7. Reporting	Send values and status to the configured output interface.	Human-readable output

Slope Detection

Slope detection is included as a monitoring feature in the project. The exact slope thresholds and decision logic should be kept synchronized with the final firmware source when the system is modified.

```

COM15 - PuTTY
pH=7.16 TDS=807.79 Turb=1.97 | SAFE
pH=7.33 TDS=786.79 Turb=2.05 | SAFE
pH=7.10 TDS=805.40 Turb=1.99 | SAFE
pH=7.20 TDS=795.79 Turb=2.04 | SAFE
pH=7.30 TDS=805.40 Turb=2.09 | SAFE
pH=7.05 TDS=788.29 Turb=1.91 | SAFE
pH=7.25 TDS=799.09 Turb=1.94 | SAFE
pH=7.06 TDS=799.70 Turb=1.99 | SAFE
[POWER] System nominal - low power mode
pH=7.32 TDS=792.20 Turb=2.05 | SAFE
pH=7.08 TDS=796.40 Turb=2.09 | SAFE
pH=7.29 TDS=790.09 Turb=1.97 | SAFE
[DETECT] Slope trigger
[DETECT] Anomaly score=0.08
[STATE] ALERT - boosting priorities
pH=3.47 TDS=5199.10 Turb=6.25 | ALERT [slope]
[DETECT] Slope trigger
[DETECT] Anomaly score=0.09
pH=3.47 TDS=5185.89 Turb=6.38 | ALERT [slope]
[DETECT] Slope trigger
[DETECT] Anomaly score=0.08
[STATE] CRITICAL - SOS
pH=3.49 TDS=5202.70 Turb=6.24 | CRITICAL [slope]
[DETECT] Slope trigger
  
```

8. Dataset & TinyML Pipeline

Dataset Structure

Column	Role
pH	Model input feature 1
TDS	Model input feature 2
turbidity	Model input feature 3
potability	Target / label column

Dataset file: water_potability.csv

Total samples supplied: 3,276

Model inputs: 3 features (pH, TDS, turbidity); potability is the fourth dataset column/target.

Train/Test Split

The project specification supplied for this manual states an 80% training / 20% test split. A separate note supplied 2,620 / 1,106 as split counts; those counts sum to 3,726 rather than 3,276. For a formal submission, verify the exact split counts in the training script or dataset log before printing a final signed copy.

TinyML Framework

Training tool: TensorFlow

Autoencoder Architecture

$3 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 3$

- Input layer: 3 normalized sensor features.
- Encoder: $3 \rightarrow 8 \rightarrow 4 \rightarrow 2$.
- Latent representation: 2 neurons.
- Decoder: $2 \rightarrow 4 \rightarrow 8 \rightarrow 3$.
- Generated weights and biases are compiled into firmware as static floating-point arrays.

9. Normalization & Anomaly Detection Parameters

Normalization Parameters

Feature	Minimum	Maximum
pH	0.22749905	13.17540172
TDS	728.75082960	56488.67241000
Turbidity	1.49220661	6.49424947

The normalization header supplied for the firmware defines these constants as PH_MIN/PH_MAX, TDS_MIN/TDS_MAX and TURBIDITY_MIN/TURBIDITY_MAX.

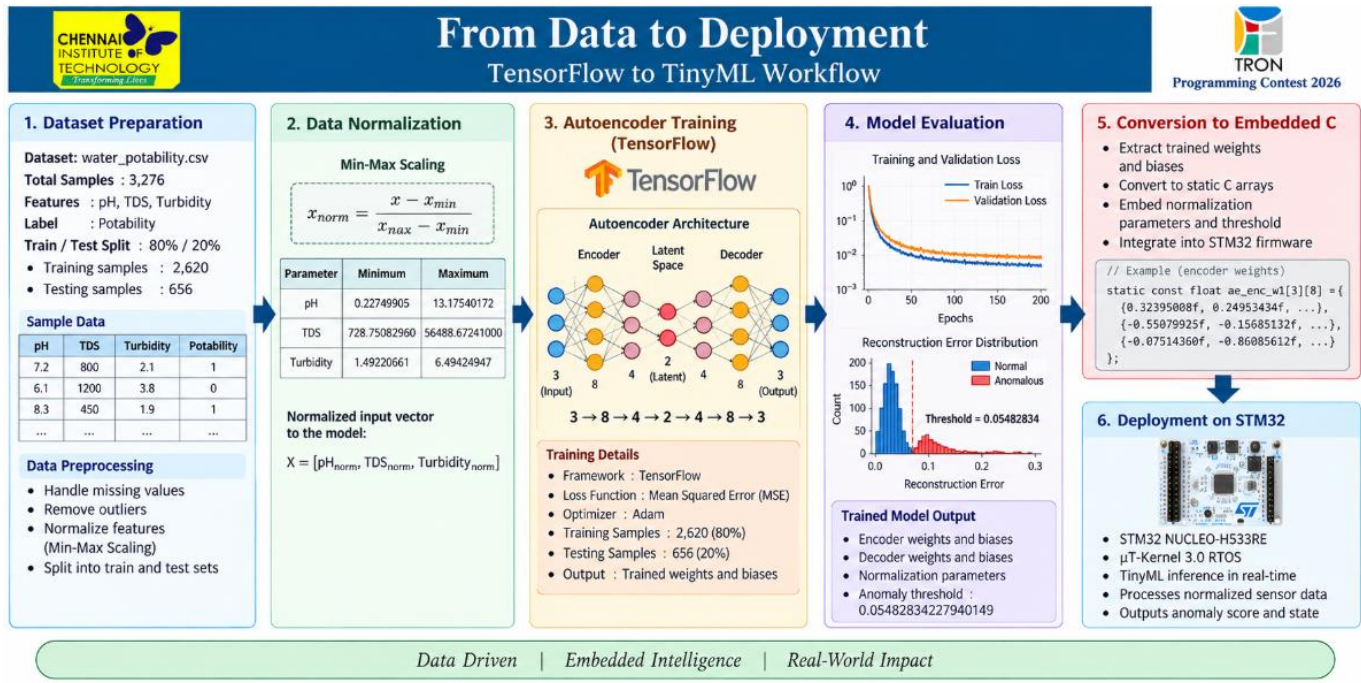
Anomaly Threshold

Reconstruction-error threshold: 0.05482834227940149

The autoencoder reconstructs the three-feature input vector. A reconstruction error is calculated and compared with the stored threshold to determine whether the sample is treated as anomalous by the embedded decision logic.

Embedded Model Parameters

The final firmware contains the generated encoder/decoder weights and biases as static const float arrays. For maintainability, the source code should be treated as the authoritative record of the complete numeric weight matrices. The architecture and normalization values are reproduced in this manual for hardware/software handover.



10. Operating Procedure — Quick Start

17. Reassemble the Nucleo, breadboard, three sensor modules and probes using the wiring tables in Section 4.
18. Confirm that all sensor VCC lines are on the Nucleo 5V rail.
19. Confirm that all sensor grounds share the Nucleo GND.
20. Confirm pH OUT passes through $R1 = 10\text{ k}\Omega$ to A0, with $R2 = 10\text{ k}\Omega$ from A0 to GND.
21. Confirm TDS analog output is connected to A1.
22. Confirm turbidity analog output is connected to A5.
23. Connect the programming/debug interface and power the controller.
24. Open the configured serial terminal and verify startup/output messages.
25. Place the probes in the prepared water sample.
26. Wait for stable readings and observe pH, TDS, turbidity and anomaly status.
27. Record results if required for the competition demonstration.

Expected First-Level Electrical Checks

Check	Expected relationship
Sensor VCC to sensor GND	Approximately the module supply voltage when powered
pH raw OUT to GND	Sensor-dependent; must remain within the divider/input design assumptions
pH A0 node to GND	Approximately half the pH OUT for equal $10\text{ k}\Omega$ resistors, under ideal conditions
TDS analog OUT to GND	Sensor-dependent analog value
Turbidity analog OUT to GND	Sensor-dependent analog value

11. Calibration & Verification Checklist

Electrical Verification

Item	Pass condition	Checked
Common ground	All three modules share Nucleo GND	☐
5 V supply	All three modules connected to Nucleo 5V	☐
pH R1	10 k Ω between pH OUT and A0 node	☐
pH R2	10 k Ω between A0 node and GND	☐
pH ADC node	A0 connected to resistor junction	☐
TDS	Analog output connected to A1	☐
Turbidity	Analog output connected to A5	☐

Software Verification

Item	Pass condition	Checked
Dataset	water_potability.csv available in training workflow	☐
Normalization	Firmware constants match Section 9	☐
Architecture	3-8-4-2-4-8-3	☐
Threshold	0.05482834227940149	☐
Serial output	Values and status visible	☐
Slope detection	Final firmware behavior verified	☐

12. Troubleshooting Guide

Symptom	Likely check	Action
A0 reads near the full sensor output	pH divider bypassed or miswired	Verify pH OUT → R1 → A0 and A0 → R2 → GND.
A0 reads 0 V	Open ground, open resistor path, wrong node or disconnected wire	Check continuity from A0 to R1/R2 junction and common GND.
TDS output reads 0 V	Wrong output pin or missing common ground	Verify module VCC/GND and analog output pin; confirm A1 connection.
Turbidity output reads 0 V	Wrong output pin, wiring fault or module configuration	Verify VCC/GND/OUT and inspect the sensor module's output circuitry.
ADC saturates at maximum	Input above ADC range or wrong wiring	Power down and verify the signal-conditioning network before reconnecting.
Serial output absent	Debug/USB connection or firmware startup issue	Verify cable, terminal settings and firmware programming.
Values fluctuate strongly	Loose wiring, electrical noise or unstable sample	Secure wiring, use common ground and allow sensor readings to stabilize.
Anomaly status unexpected	Normalization or model parameters differ from training	Verify min/max constants, model arrays and threshold against the final source.

Important Diagnostic Rule

When troubleshooting, power down before moving jumper wires or resistor connections. Never use continuity mode on a powered circuit.

13. Dismantling, Transport & Reassembly

Dismantling for Shipment

28. Power off and disconnect USB/programming power.
29. Remove probes/sensor electrodes from any liquid and dry them according to their storage requirements.
30. Photograph the complete assembled wiring before disconnecting anything.
31. Label the pH, TDS and turbidity module connectors/wires.
32. Keep R1 and R2 together with the pH signal wiring; do not discard or substitute them.
33. Disconnect sensor power and signal wires carefully.
34. Protect the Nucleo board, sensor modules and probes from electrostatic discharge, moisture and mechanical impact.
35. Pack each sensor/probe separately where appropriate.

Reassembly Order

36. Mount the Nucleo and breadboard securely.
37. Establish the common GND rail.
38. Connect 5V to the three modules.
39. Build and verify the pH divider.
40. Connect TDS output to A1 and turbidity output to A5.
41. Inspect for accidental shorts or crossed signal wires.
42. Power the system and perform the electrical verification checklist before placing probes in water.

14. Safety, Handling & Maintenance

- Keep the Nucleo board, breadboard, USB connector and exposed electronics dry.
- Only the sensor/probe portions intended for immersion should enter the water sample.
- Disconnect power before changing wiring or resistor connections.
- Do not perform continuity tests on an energized circuit.
- Use the sensor manufacturer's cleaning and storage procedure for the pH electrode and other probes.
- Do not bend, impact or contaminate probe tips/electrodes.
- Keep the pH divider resistors at the specified 10 k Ω values unless the firmware/electrical design is intentionally recalculated.
- Before every demonstration, perform a visual inspection and a short electrical sanity check.

Maintenance Log

Date	Action / observation	Performed by

15. Technical Quick Reference

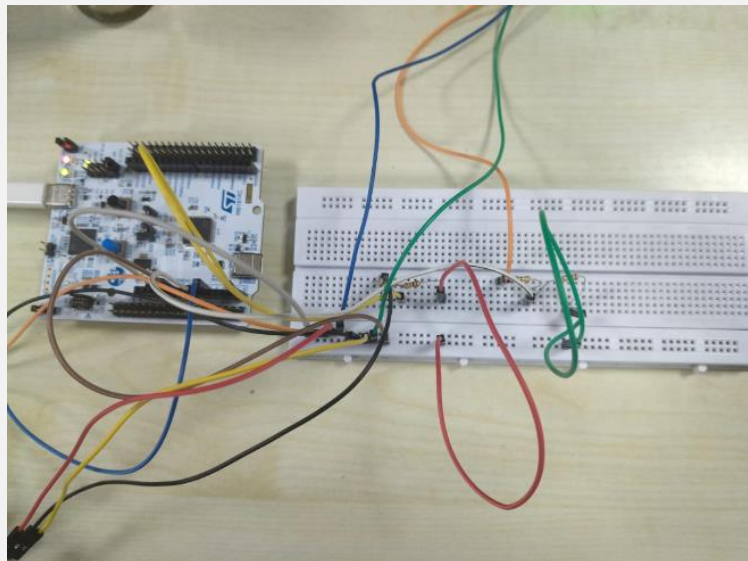
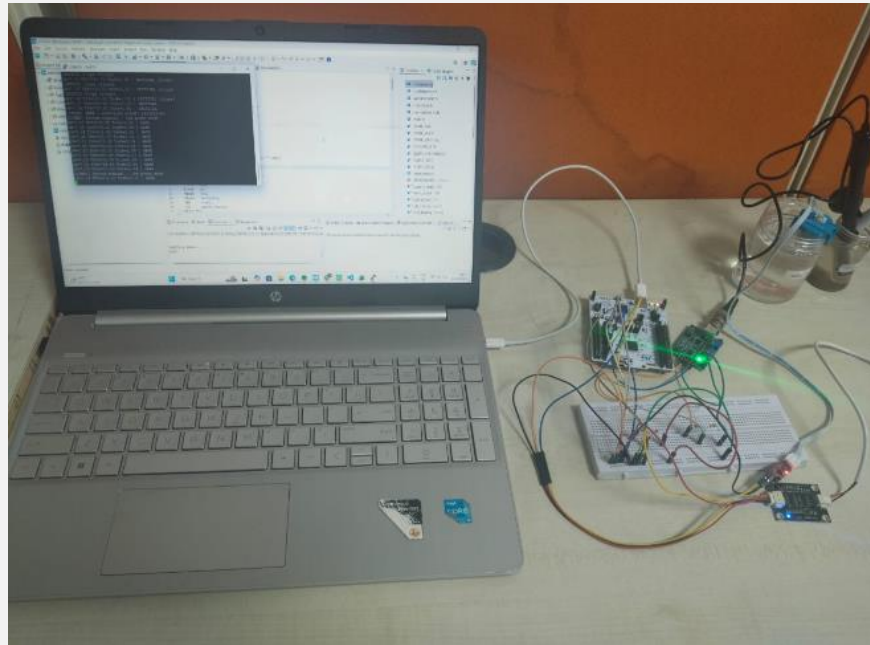
Parameter	Value
Controller ADC channels	A0 = pH, A1 = TDS, A5 = turbidity
Sensor supply	5 V
Common ground	Yes — all three modules to Nucleo GND
pH divider R1	10 k Ω , pH OUT → A0 node
pH divider R2	10 k Ω , A0 node → GND
Dataset	water_potability.csv
Dataset samples	3,276
Dataset columns	4: pH, TDS, turbidity, potability
Model input features	3
Autoencoder	3 → 8 → 4 → 2 → 4 → 8 → 3
Training tool	TensorFlow
Normalization	Feature-wise min-max using stored constants
Anomaly threshold	0.05482834227940149
Slope detection	Included in project specification; verify final firmware thresholds/logic

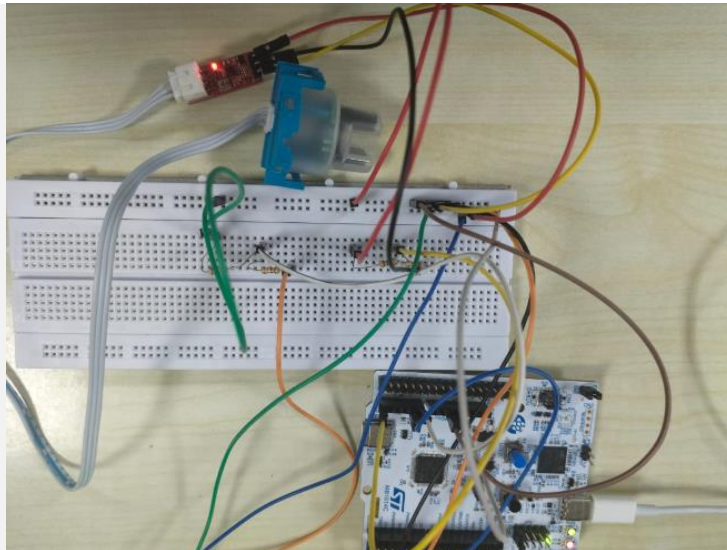
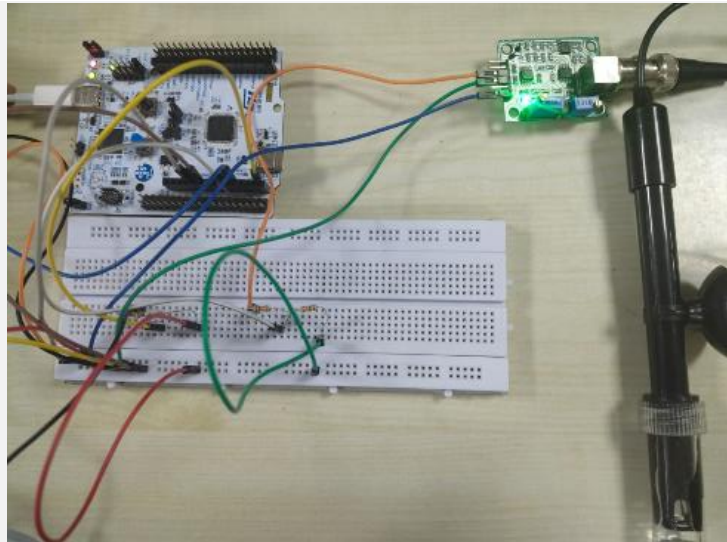
Source-Code Handover

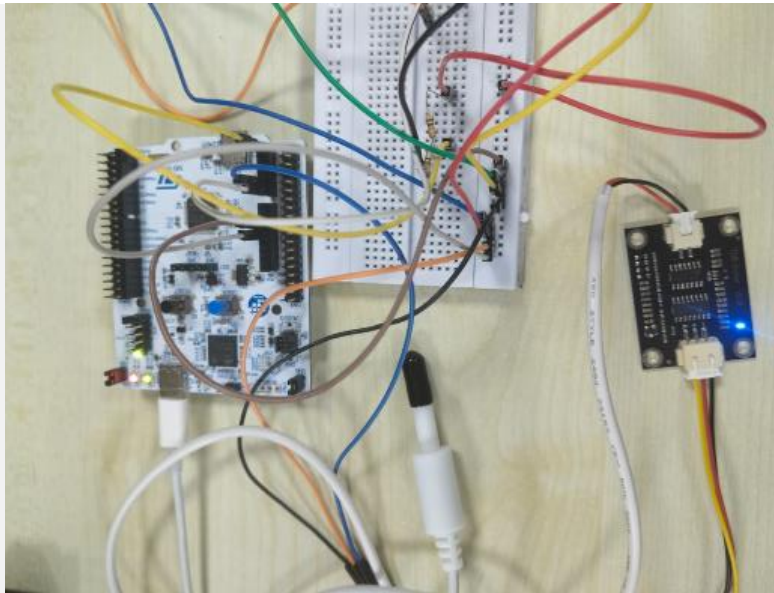
The hardware should be transported together with the final firmware/source package. The source package is the authoritative reference for exact RTOS task names, timing values, ADC configuration, conversion formulas, complete autoencoder weights/biases, serial settings and any slope-detection thresholds not explicitly documented in this manual.

16. Figure & Photo Register

The following spaces are intentionally reserved so the team can insert real project photographs before printing.







```
COM15 - PuTTY
pH=7.16 TDS=807.79 Turb=1.97 | SAFE
pH=7.33 TDS=786.79 Turb=2.05 | SAFE
pH=7.10 TDS=805.40 Turb=1.99 | SAFE
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pH=7.30 TDS=805.40 Turb=2.09 | SAFE
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[DETECT] Slope trigger
[DETECT] Anomaly score=0.08
[STATE] CRITICAL - SOS
pH=3.49 TDS=5202.70 Turb=6.24 | CRITICAL [slope]
[DETECT] Slope trigger
```

17. Handover Sign-Off

This page may be signed when the hardware, firmware and documentation have been checked together before shipment.

Handover item	Status
Hardware wiring verified	☐ Complete ☐ Pending
pH divider verified	☐ Complete ☐ Pending
TDS channel A1 verified	☐ Complete ☐ Pending
Turbidity channel A5 verified	☐ Complete ☐ Pending
Firmware programmed	☐ Complete ☐ Pending
Serial output verified	☐ Complete ☐ Pending
TinyML model verified	☐ Complete ☐ Pending
Photos inserted into manual	☐ Complete ☐ Pending
Source code package included	☐ Complete ☐ Pending
Shipment packing verified	☐ Complete ☐ Pending

Role	Name	Signature	Date
Team representative	_____	_____	_____
Faculty / mentor	_____	_____	_____
Receiver / handover	_____	_____	_____